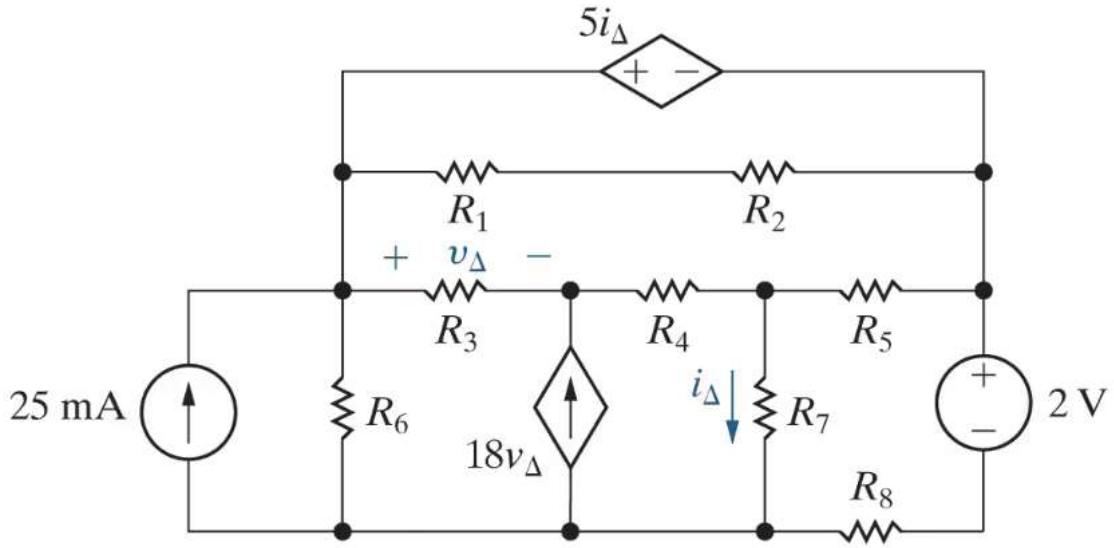
Problems - Week 5

- Node-Voltage Method
- Mesh-Current Method

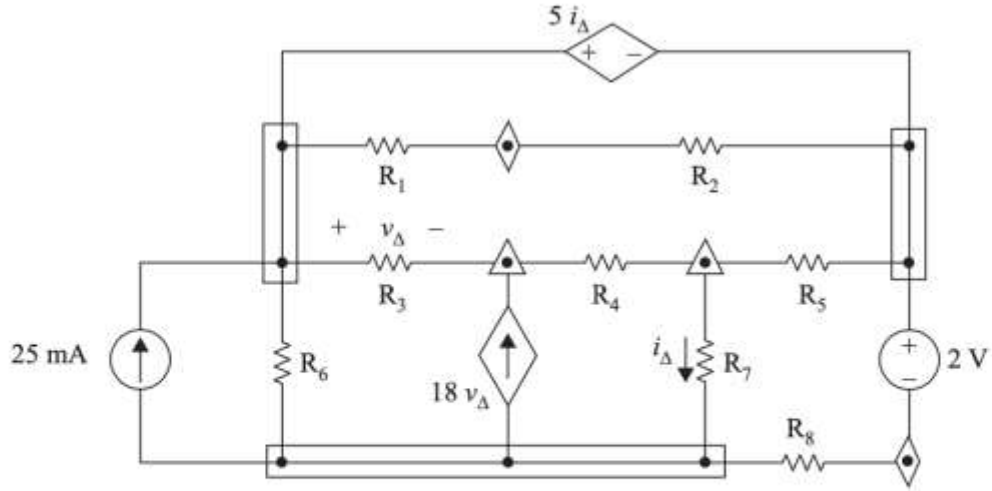
Identifying Node, Branch, Mesh, and Loop in a Circuit

4.1 For the circuit shown in Fig. P4.1, state the numerical value of the number of (a) branches, (b) branches where the current is unknown, (c) essential branches, (d) essential branches where the current is unknown, (e) nodes, (f) essential nodes, and (g) meshes.

Figure P4.1



P 4.1



[a] 12 branches, 8 branches with resistors, 2 branches with independent sources, 2 branch with dependent source.

[b] The current is unknown in every branch except the one containing the 25 mA current source, so the current is unknown in 11 branches.

[c] 10 essential branches: $-R_1 - R_2$ forms an essential branch as does $R_8 - 2$ V. The remaining eight branches are essential branches that contain a single element.

[d] The current is known only in the essential branch containing the current source and is unknown in the remaining 9 essential branches.

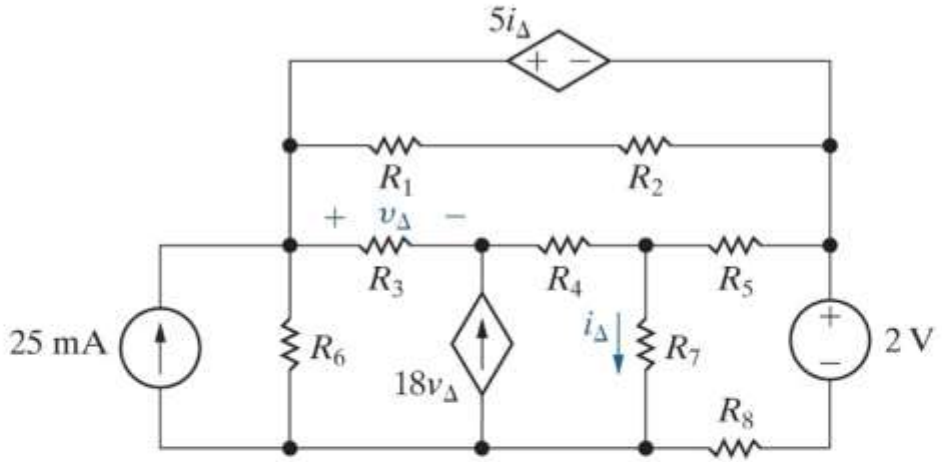
[e] From the figure there are 7 nodes - three identified by rectangular boxes, two identified by triangles, and two identified by diamonds.

[f] There are 5 essential nodes, three identified with rectangular boxes and two identified with triangles.

[g] A mesh is like a windowpane, and as can be seen from the figure there are 6 windowpanes or meshes.

Identifying Node, Branch, Mesh, and Loop in a Circuit

Figure P4.1



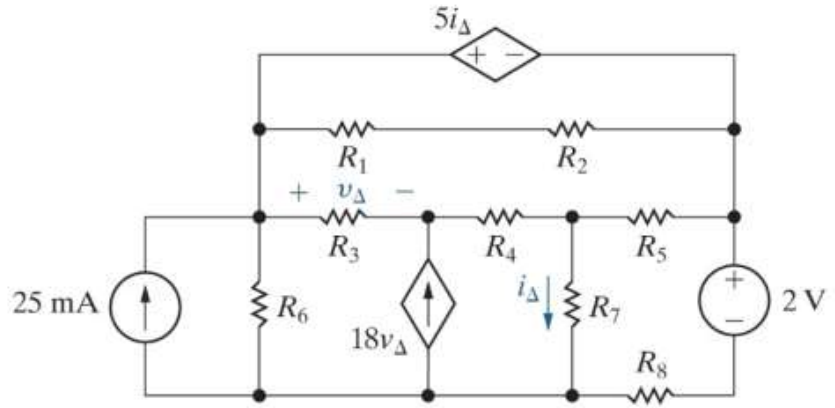
4.2 **a.** If only the essential nodes and branches are identified in the circuit in Fig. P4.1, how many simultaneous equations are needed to describe the circuit?

b. How many of these equations can be derived using Kirchhoff's current law?

c. How many must be derived using Kirchhoff's voltage law?

Identifying Node, Branch, Mesh, and Loop in a Circuit

Figure P4.1



4.2 **a.** If only the essential nodes and branches are identified in the circuit in Fig. P4.1, how many simultaneous equations are needed to describe the circuit?

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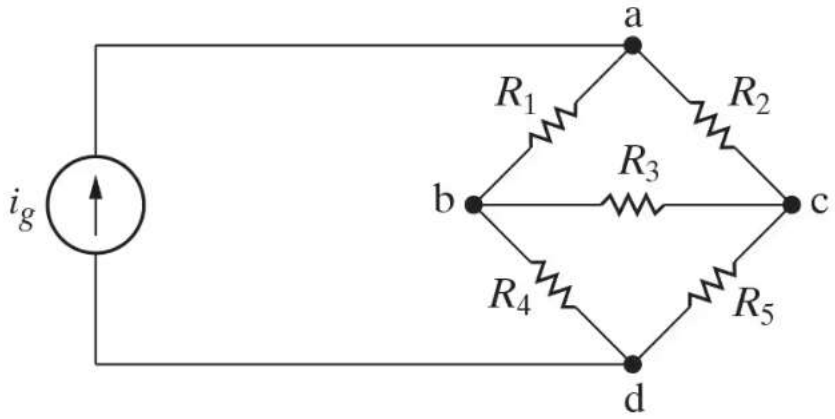
P 4.2 **[a]** From Problem 4.1(d) there are 10 essential branches where the current is unknown, so we need 10 simultaneous equations to describe the circuit.

[b] From Problem 4.1(f), there are 5 essential nodes, so we can apply KCL at $(5-1)=4$ of these essential nodes. There would also be two dependent source constraint equations.

[c] The remaining 4 equations needed to describe the circuit will be derived from KVL equations.

Identifying Node, Branch, Mesh, and Loop in a Circuit

4.3 Assume the current i_g in the circuit in Fig. P4.3 is known.



a. How many unknown currents are there?

b. How many independent equations can be written using Kirchhoff's current law (KCL)?

c. Write an independent set of KCL equations.

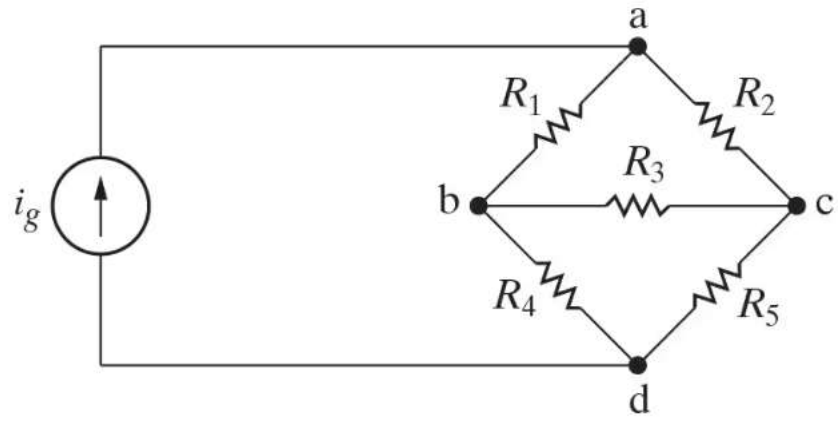
d. How many independent equations can be derived from Kirchhoff's voltage law (KVL)?

e. Write a set of independent KVL equations.

[a] There are six circuit components, five resistors and the current source. Since the current is known only in the current source, it is unknown in the five resistors. Therefore there are **five** unknown currents.

[b] There are four essential nodes in this circuit, identified by the dark black dots in Fig. P4.3. At three of these nodes you can write KCL equations that will be independent of one another. A KCL equation at the fourth node would be dependent on the first three. Therefore there are **three** independent KCL equations.

4.3 Assume the current i_g in the circuit in Fig. P4.3 is known.



a. How many unknown currents are there?

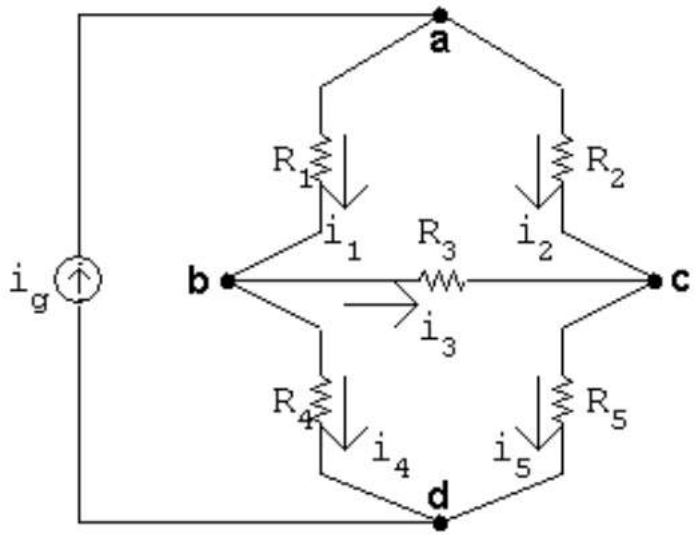
b. How many independent equations can be written using Kirchhoff's current law (KCL)?

c. Write an independent set of KCL equations.

d. How many independent equations can be derived from Kirchhoff's voltage law (KVL)?

e. Write a set of independent KVL equations.

[c]



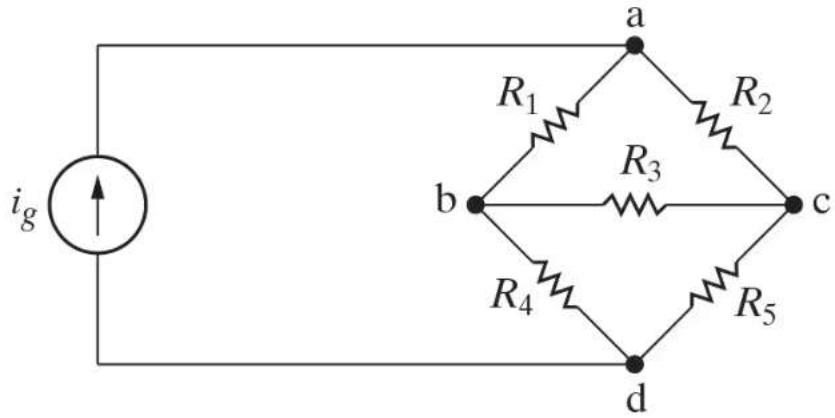
Sum the currents at any three of the four essential nodes a, b, c, and d. Using nodes a, b, and c we get

$$-i_g + i_1 + i_2 = 0;$$

$$-i_1 + i_4 + i_3 = 0;$$

$$i_5 - i_2 - i_3 = 0.$$

4.3 Assume the current i_g in the circuit in Fig. P4.3 is known.



a. How many unknown currents are there?

b. How many independent equations can be written using Kirchhoff's current law (KCL)?

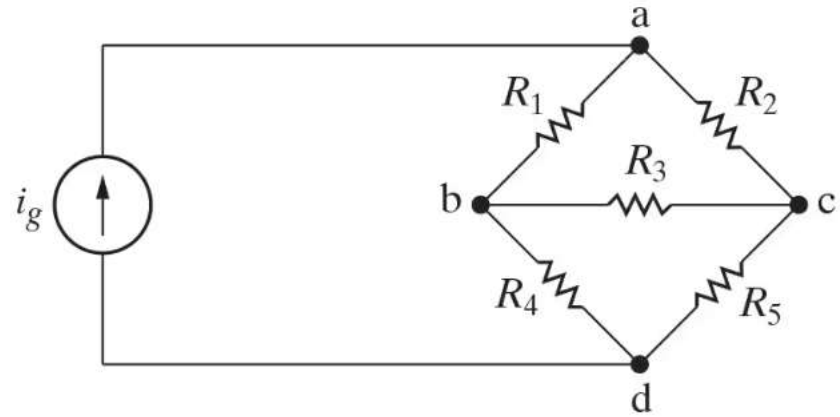
c. Write an independent set of KCL equations.

d. How many independent equations can be derived from Kirchhoff's voltage law (KVL)?

e. Write a set of independent KVL equations.

[d] There are three meshes in this circuit: one on the left with the components i_g, R_1, and R_4; one on the top right with components R_1, R_2, and R_3; and one on the bottom right with components R_3, R_4, and R_5. We cannot write a KVL equation for the left mesh because we don't know the voltage drop across the current source. Therefore, we can write KVL equations for the two meshes on the right, giving a total of **two** independent KVL equations.

4.3 Assume the current i_g in the circuit in Fig. P4.3 is known.



a. How many unknown currents are there?

b. How many independent equations can be written using Kirchhoff's current law (KCL)?

c. Write an independent set of KCL equations.

d. How many independent equations can be derived from Kirchhoff's voltage law (KVL)?

e. Write a set of independent KVL equations.

[e] Sum the voltages around two independent closed paths, avoiding a path that contains the independent current source since the voltage across the current source is not known. Using the upper and lower meshes formed by the five resistors gives

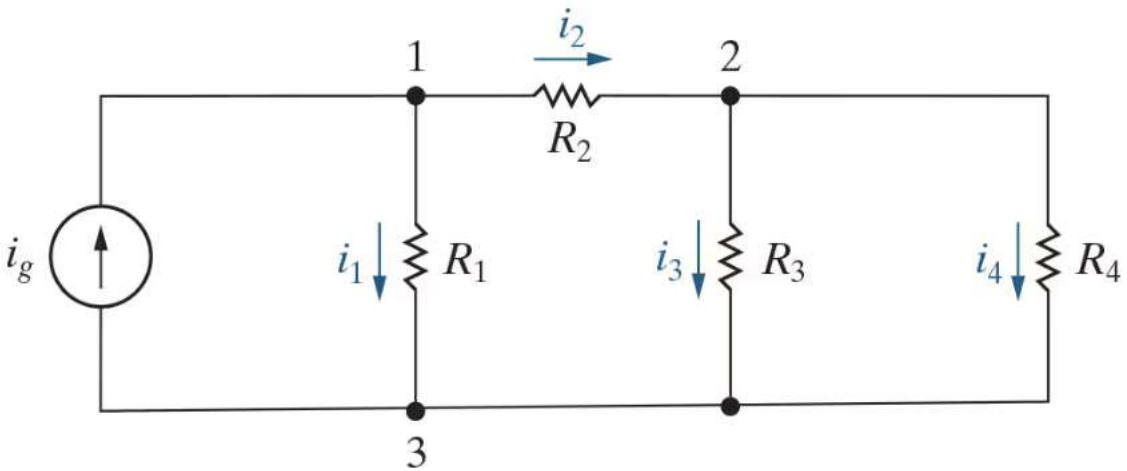
$$R_1 i_1 + R_3 i_3 - R_2 i_2 = 0;$$

$$R_3 i_3 + R_5 i_5 - R_4 i_4 = 0.$$

4.5 A current leaving a node is defined as positive.

a. Sum the currents at each essential node in the circuit shown in Fig. P4.5.

b. Show that any one of the equations in (a) can be derived from the remaining three equations.



[a] At node 1: $-i_g + i_1 + i_2 = 0;$

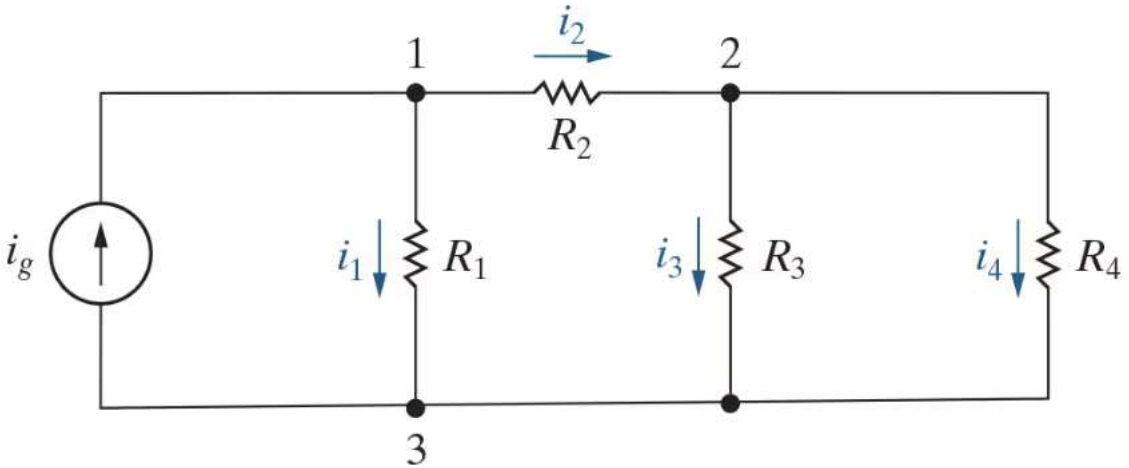
At node 2: $-i_2 + i_3 + i_4 = 0;$

At node 3: $i_g - i_1 - i_3 - i_4 = 0.$

4.5 A current leaving a node is defined as positive.

a. Sum the currents at each essential node in the circuit shown in Fig. P4.5.

b. Show that any one of the equations in (a) can be derived from the remaining three equations.



[a] At node 1: $\quad -i_g + i_1 + i_2 = 0;$

At node 2: $\quad -i_2 + i_3 + i_4 = 0;$

At node 3: $\quad i_g - i_1 - i_3 - i_4 = 0.$

[b] There are many possible solutions. For example, solve the equation at node 1 for i_g:

$$i_g = i_1 + i_2.$$

Substitute this expression for i_g into the equation at node 3:

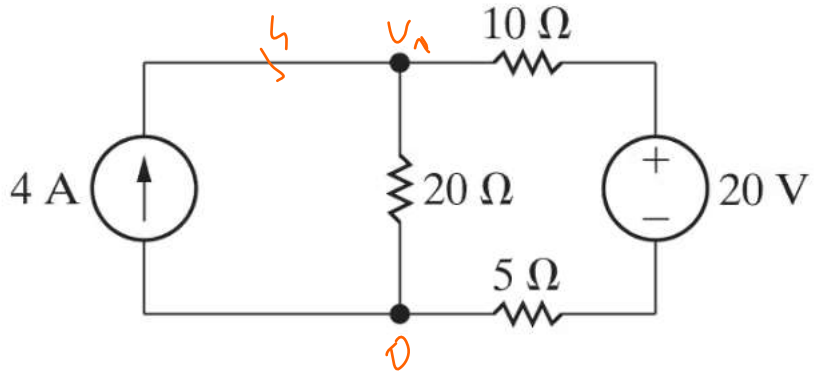
$$(i_1 + i_2) - i_1 - i_3 - i_4 = 0 \qquad \text{so} \qquad i_2 - i_3 - i_4 = 0.$$

Multiply this last equation by -1 to get the equation at node 2:

$$-(i_2 - i_3 - i_4) = -0 \qquad \text{so} \qquad -i_2 + i_3 + i_4 = 0.$$

4.6 Use the node-voltage method to find how much power the 4 A source extracts from the circuit in Fig. P4.6.

Figure P4.6

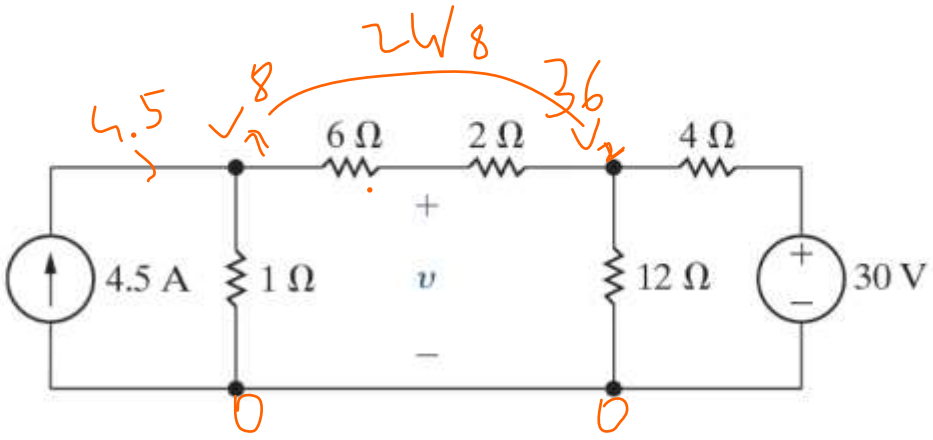


P 4.6 $$-4+\frac{v_o}{20}+\frac{v_o-20}{10+5}=0 \Rightarrow v_0 = 45.7142\text{V}$$

$$p_{4\text{A}} = -(45.714)(4) = -182.857\text{W} \qquad \text{(delivering)}.$$

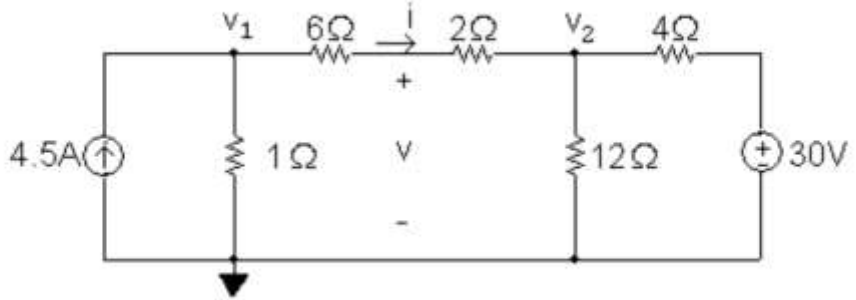
4.7 Use the node-voltage method to find v in the circuit in Fig. P4.7.

Figure P4.7



4.7 Use the node-voltage method to find v in the circuit in Fig. P4.7.

P 4.7 Redraw the circuit, choosing the node voltages and reference node as shown:



The two node voltage equations are:

$$-4.5 + \frac{v_1}{1} + \frac{v_1 - v_2}{6+2} = 0;$$

$$\frac{v_2}{12} + \frac{v_2 - v_1}{6+2} + \frac{v_2 - 30}{4} = 0.$$

Place these equations in standard form:

$$v_1\left(1 + \frac{1}{8}\right) + v_2\left(-\frac{1}{8}\right) = 4.5;$$

$$v_1\left(-\frac{1}{8}\right) + v_2\left(\frac{1}{12} + \frac{1}{8} + \frac{1}{4}\right) = 7.5.$$

Solving, $v_1 = 6$ V $\qquad v_2 = 18$ V.

To find the voltage v, first find the current i through the series-connected $6\,\Omega$ and $2\,\Omega$ resistors:

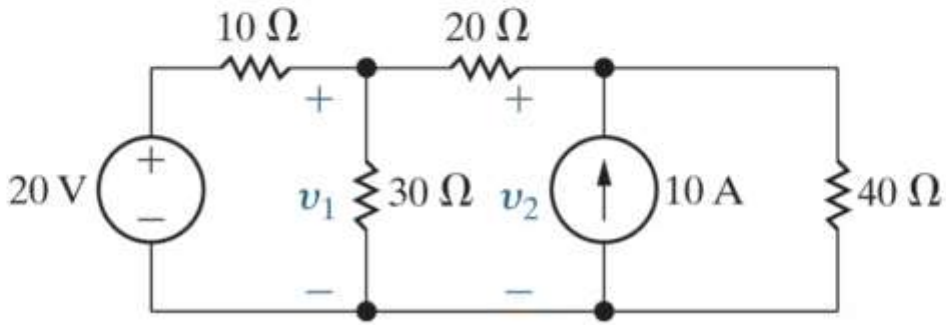
$$i = \frac{v_1 - v_2}{6+2} = \frac{6 - 18}{8} = -1.5\,\text{A}.$$

Using a KVL equation, calculate v:

$$v = 2i + v_2 = 2(-1.5) + 18 = 15\,\text{V}.$$

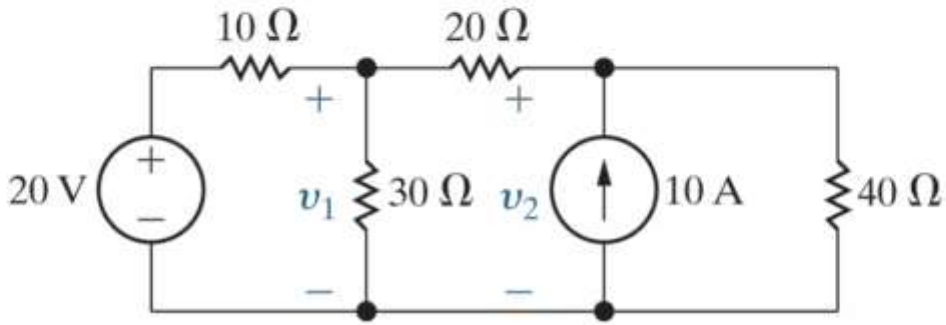
4.13 Use the node-voltage method to find v_1 and v_2 in the circuit in Fig. P4.13.

Figure P4.13



4.13 Use the node-voltage method to find v_1 and v_2 in the circuit in Fig. P4.13.

Figure P4.13



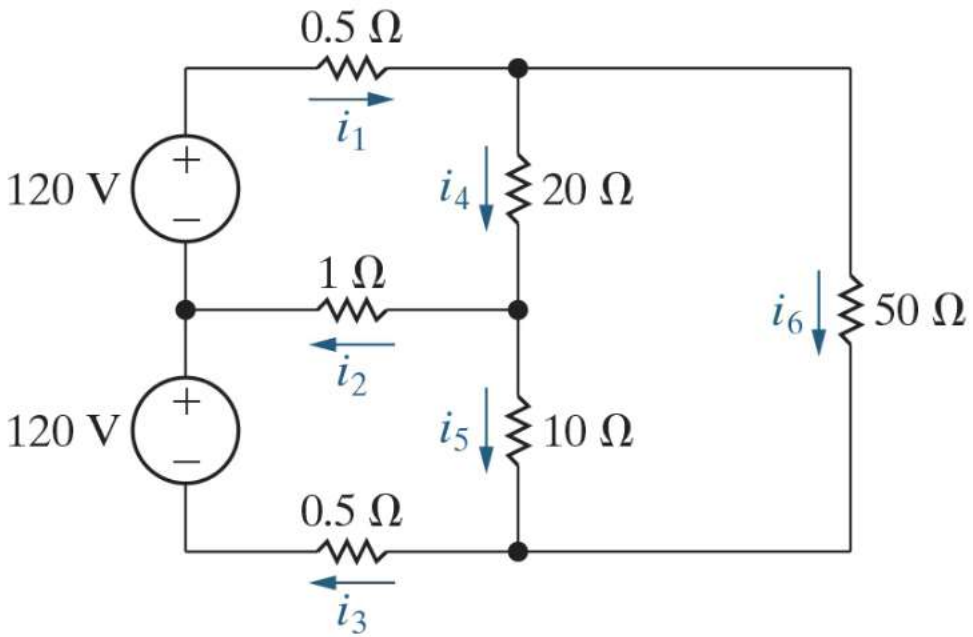
$$\frac{v_1 - 20}{10} + \frac{v_1}{30} + \frac{v_1 - v_2}{20} = 0 \qquad \text{so} \qquad 11v_1 - 3v_2 = 120;$$

$$-10 + \frac{v_2 - v_1}{20} + \frac{v_2}{40} = 0 \qquad \text{so} \qquad -0.05v_1 + 0.075v_2 = 10.$$

Solving, $v_1 = 57.77$ V; $\quad v_2 = 171.85$ V.

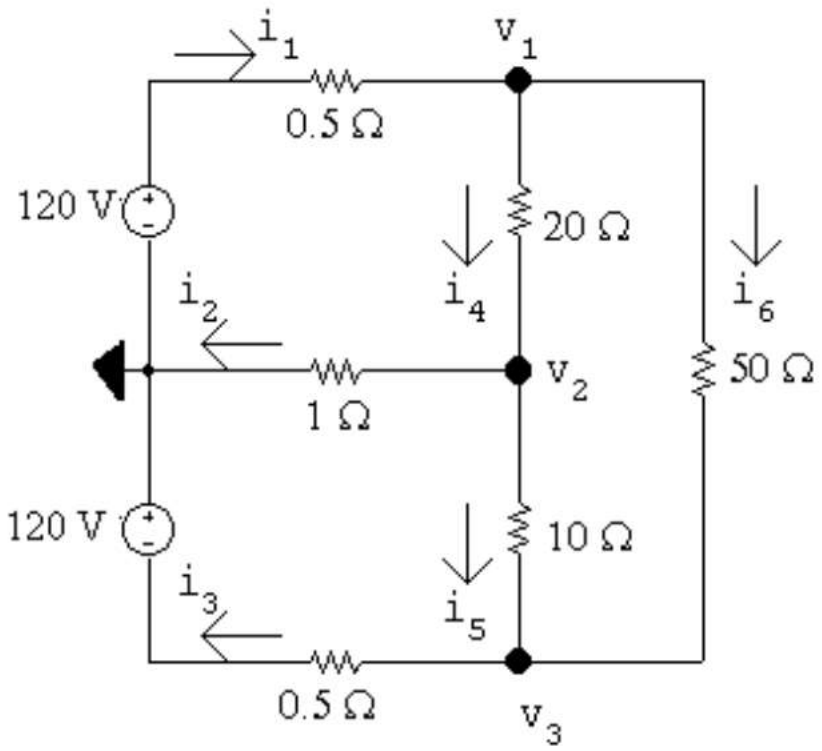
4.15 The circuit shown in Fig. P4.15 is a dc model of a residential power distribution circuit.

a. Use the node-voltage method to find the branch currents $i_1 - i_6$.



4.15 The circuit shown in Fig. P4.15 is a dc model of a residential power distribution circuit.

a. Use the node-voltage method to find the branch currents i_1–i_6.



$$\frac{v_1 - 120}{0.5} + \frac{v_1 - v_2}{20} + \frac{v_1 - v_3}{50} = 0;$$
$$\frac{v_2 - v_1}{20} + \frac{v_2}{1} + \frac{v_2 - v_3}{10} = 0;$$
$$\frac{v_3 + 120}{0.5} + \frac{v_3 - v_2}{10} + \frac{v_3 - v_1}{50} = 0.$$

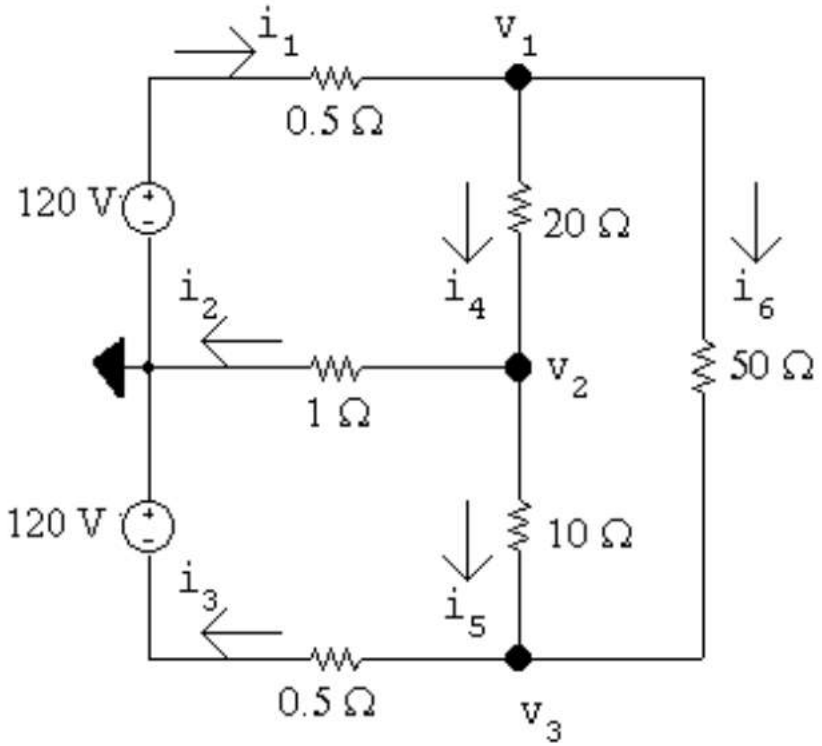
In standard form:

$$v_1\left(\frac{1}{0.5} + \frac{1}{20} + \frac{1}{50}\right) + v_2\left(-\frac{1}{20}\right) + v_3\left(-\frac{1}{50}\right) = 240;$$
$$v_1\left(-\frac{1}{20}\right) + v_2\left(\frac{1}{20} + \frac{1}{1} + \frac{1}{10}\right) + v_3\left(-\frac{1}{10}\right) = 0;$$
$$v_1\left(-\frac{1}{50}\right) + v_2\left(-\frac{1}{10}\right) + v_3\left(\frac{1}{0.5} + \frac{1}{10} + \frac{1}{50}\right) = -240.$$

Solving, $v_1 = 114.74\,\text{V}$; $v_2 = -4.78\,\text{V}$; $v_3 = -112.35\,\text{V}$.

4.15 The circuit shown in Fig. P4.15 is a dc model of a residential power distribution circuit.

a. Use the node-voltage method to find the branch currents i_1–i_6.

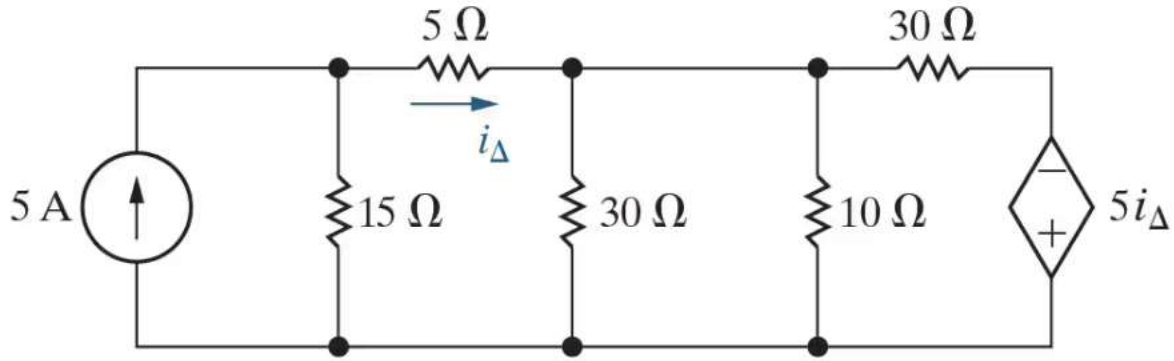


$$i_1 = \frac{120 - v_1}{0.5} = 10.52 \text{ A} \qquad i_4 = \frac{v_1 - v_2}{20} = 5.976 \text{ A};$$

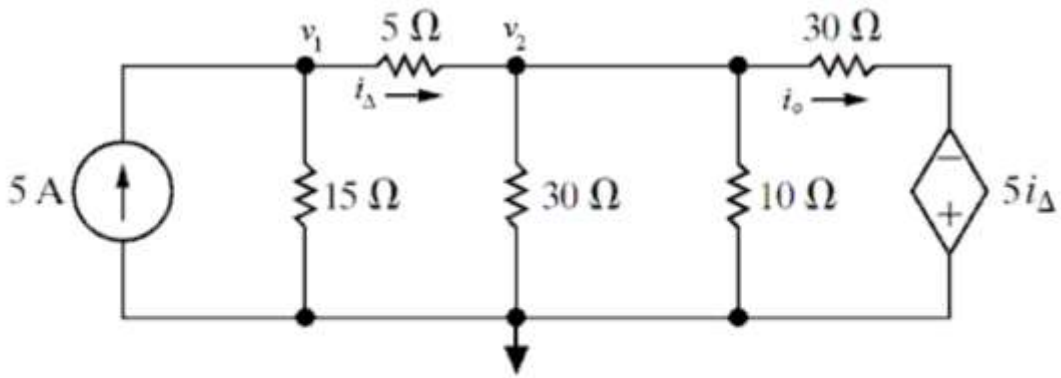
$$i_2 = \frac{v_2}{1} = -4.78 \text{ A} \qquad i_5 = \frac{v_2 - v_3}{10} = 10.757 \text{ A};$$

$$i_3 = \frac{v_3 + 120}{0.5} = 15.3 \text{ A} \qquad i_6 = \frac{v_1 - v_3}{50} = 4.542 \text{ A}.$$

4.17 **a.** Use the node-voltage method to find the total power developed in the circuit in Fig. P4.17.



4.17 **a.** Use the node-voltage method to find the total power developed in the circuit in Fig. P4.17.



Write the KCL equations and the dependent source constraint equation:

$$-5 + \frac{v_1}{15} + \frac{v_1 - v_2}{5} = 0 \quad \text{so} \quad 4v_1 - 3v_2 + 0i_\Delta = 75;$$

$$\frac{v_2 - v_1}{5} + \frac{v_2}{30} + \frac{v_2}{10} + \frac{v_2 + 5i_\Delta}{30} = 0 \quad \text{so} \quad -6v_1 + 11v_2 + 5i_\Delta = 0;$$

$$i_\Delta = \frac{v_1 - v_2}{5} \quad \text{so} \quad v_1 - v_2 - 5i_\Delta = 0.$$

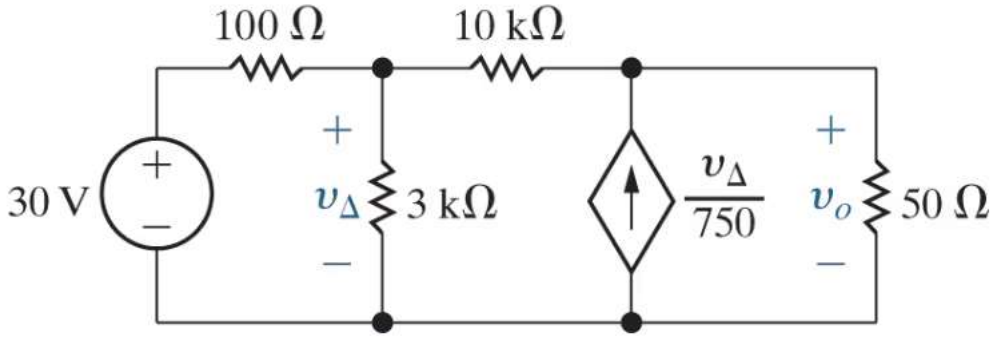
Solving, $v_1 = 30\,\text{V}$; $v_2 = 15\,\text{V}$; $i_\Delta = 3\,\text{A}$; $i_o = \dfrac{15 + 15}{30} = 1\,\text{A}$.

$$p_{5i_\Delta} = (-15)(1) = -15\,\text{W(del)};$$

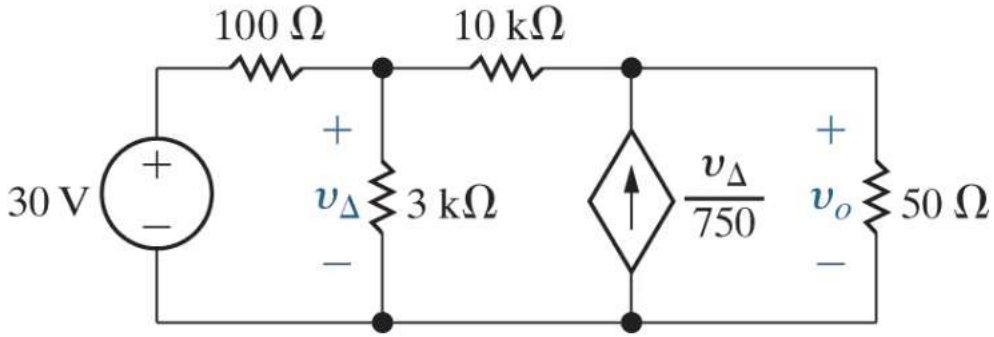
$$p_{5\text{A}} = -5(30) = -150\,\text{W(del)}$$

$$\therefore \quad p_{\text{dev}} = 165\,\text{W}.$$

4.19 **a.** Use the node-voltage method to find v_o for the circuit in Fig. P4.19.



4.19 **a.** Use the node-voltage method to find v_o for the circuit in Fig. P4.19.



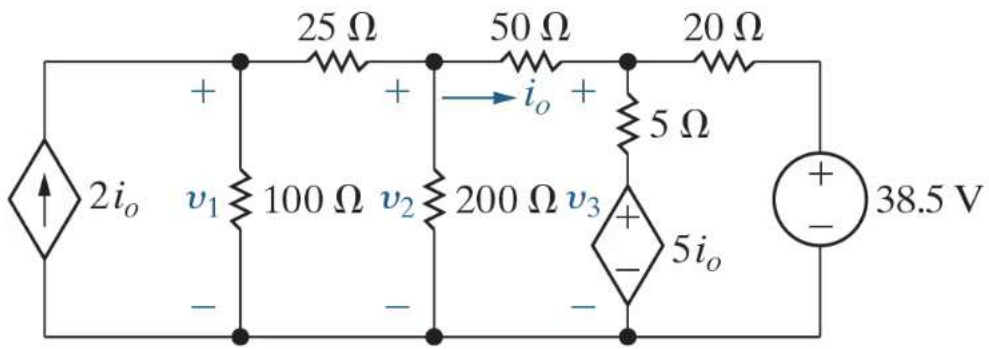
[a] $$\frac{v_\Delta - 30}{100} + \frac{v_\Delta}{3000} + \frac{v_\Delta - v_o}{10{,}000} = 0$$

&

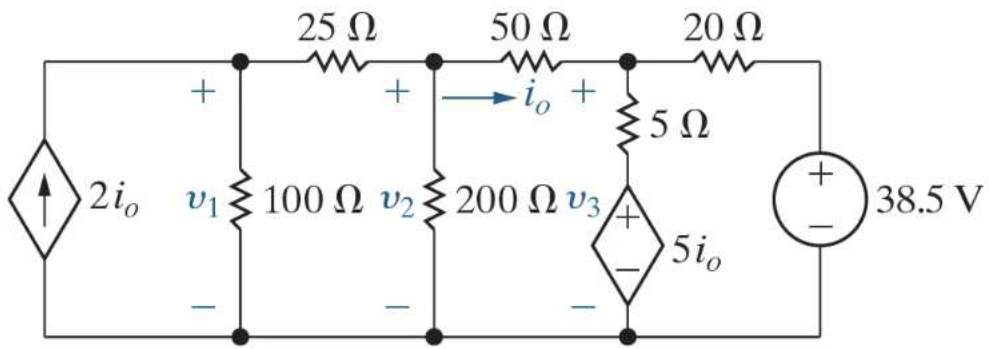
$$\frac{v_o - v_\Delta}{10{,}000} - \frac{v_\Delta}{750} + \frac{v_o}{50} = 0$$

On solving $\rightarrow v_0 = 0.2$ V; $\qquad v_\Delta = 2.8$ V. ≈ 3 V

4.20 **a.** Find the node voltages v_1, v_2, and v_3 in the circuit in Fig. P4.20.



4.20 **a.** Find the node voltages v_1, v_2, and v_3 in the circuit in Fig. P4.20.



$$i_o = \frac{v_2 - v_3}{50};$$

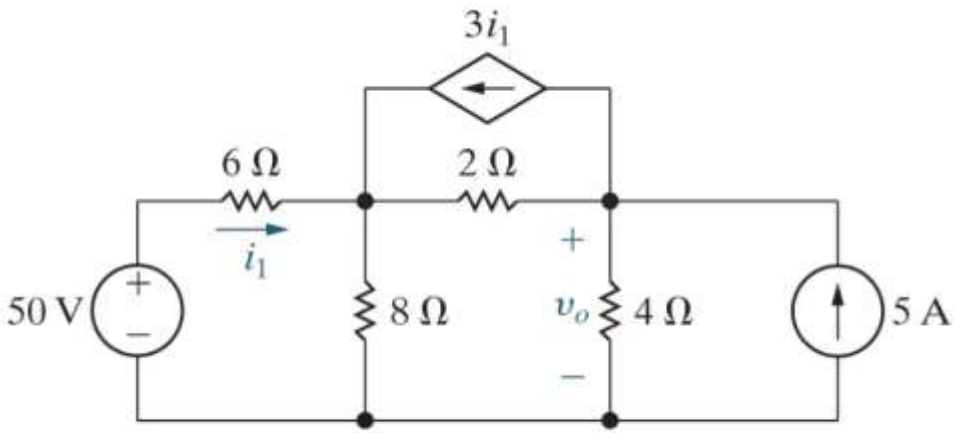
$$-2i_o + \frac{v_1}{100} + \frac{v_1 - v_2}{25} = 0 \qquad \text{so} \qquad 5v_1 - 8v_2 + 4v_3 = 0;$$

$$\frac{v_2 - v_1}{25} + \frac{v_2}{200} + \frac{v_2 - v_3}{50} = 0 \qquad \text{so} \qquad -8v_1 + 13v_2 - 4v_3 = 0;$$

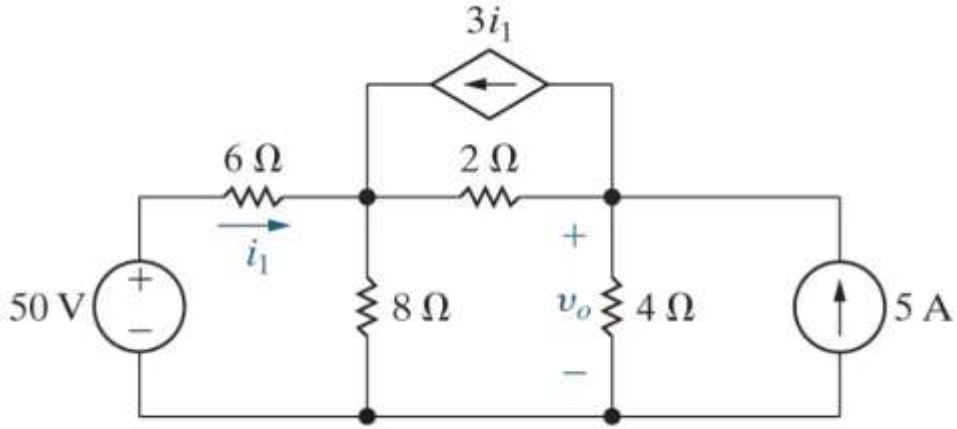
$$\frac{v_3 - v_2}{50} + \frac{v_3 - 5i_o}{5} + \frac{v_3 - 38.5}{20} = 0 \qquad \text{so} \qquad 0v_1 - 4v_2 + 29v_3 = 192.5.$$

Solving, $v_1 = -50\,\text{V}$; $v_2 = -30\,\text{V}$; $v_3 = 2.5\,\text{V}$.

4.21 **a.** Use the node-voltage method to find v_o in the circuit in Fig. P4.21.



4.21 **a.** Use the node-voltage method to find v_o in the circuit in Fig. P4.21.



The node voltage equations are:

$$\frac{v_1 - 50}{6} + \frac{v_1}{8} + \frac{v_1 - v_o}{2} - 3i_1 = 0$$

$$-5 + \frac{v_o}{4} + \frac{v_o - v_1}{2} + 3i_1 = 0.$$

The dependent source requires the following constraint equation:

$$i_1 = \frac{50 - v_1}{6}.$$

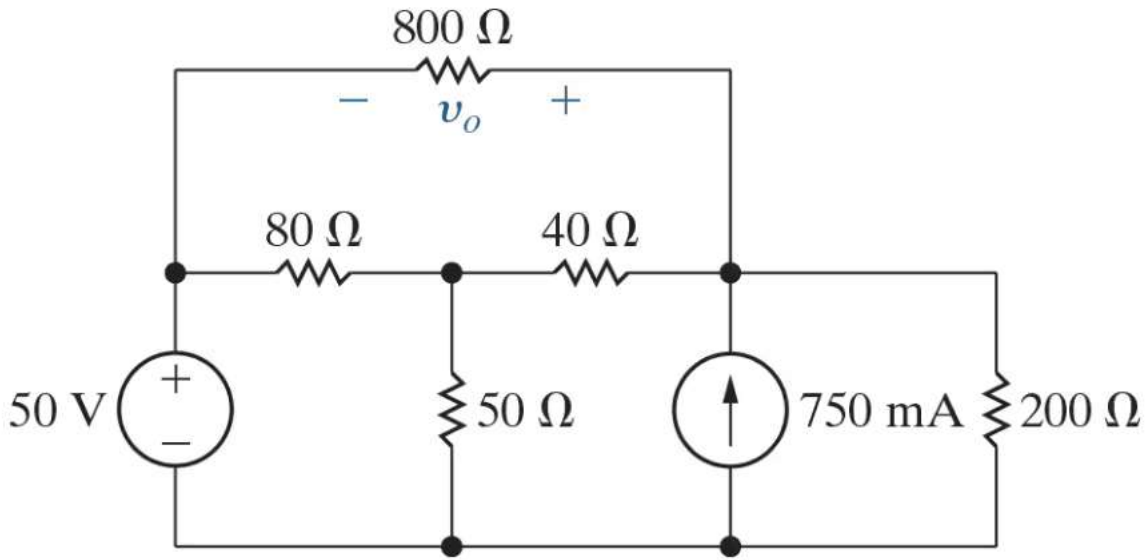
Place these equations in standard form:

$$\begin{aligned} v_1\left(\frac{1}{6} + \frac{1}{8} + \frac{1}{2}\right) &+ v_o\left(-\frac{1}{2}\right) &&+ i_1(-3) &&= \frac{50}{6} \\ v_1\left(-\frac{1}{2}\right) &+ v_o\left(\frac{1}{4} + \frac{1}{2}\right) &&+ i_1(3) &&= 5 \\ v_1\left(\frac{1}{6}\right) &+ v_o(0) &&+ i_1(1) &&= \frac{50}{6}. \end{aligned}$$

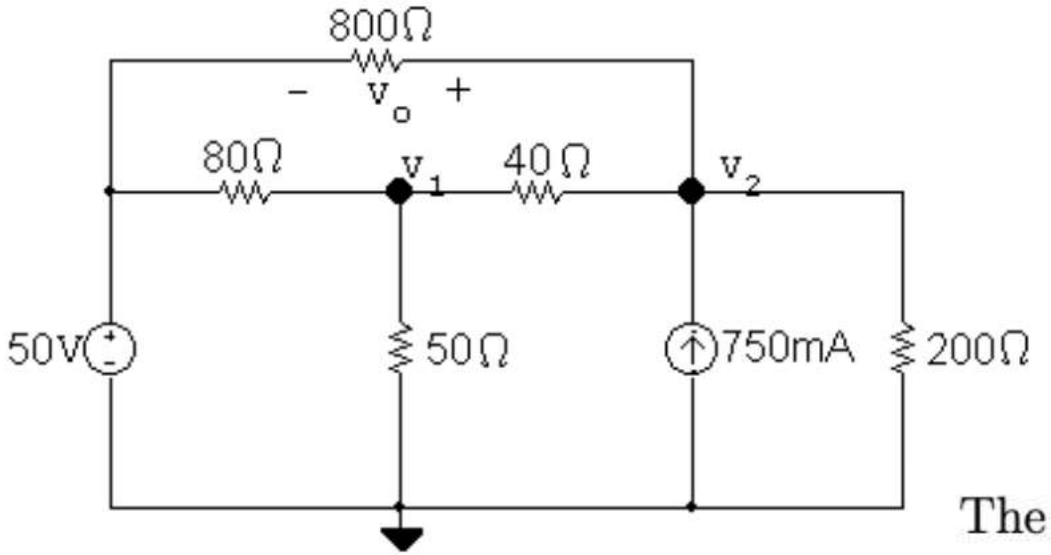
Solving, $v_1 = 32$ V; $v_o = 16$ V; $i_1 = 3$ A.

Therefore, $v_o = 16$ V.

4.25 Use the node-voltage method to find the value of v_o in the circuit in Fig. P4.25.



4.25 Use the node-voltage method to find the value of v_o in the circuit in Fig. P4.25.



The two node voltage equations are:

$$\frac{v_1 - 50}{80} + \frac{v_1}{50} + \frac{v_1 - v_2}{40} = 0;$$

$$\frac{v_2 - v_1}{40} - 0.75 + \frac{v_2}{200} + \frac{v_2 - 50}{800} = 0.$$

Place these equations in standard form:

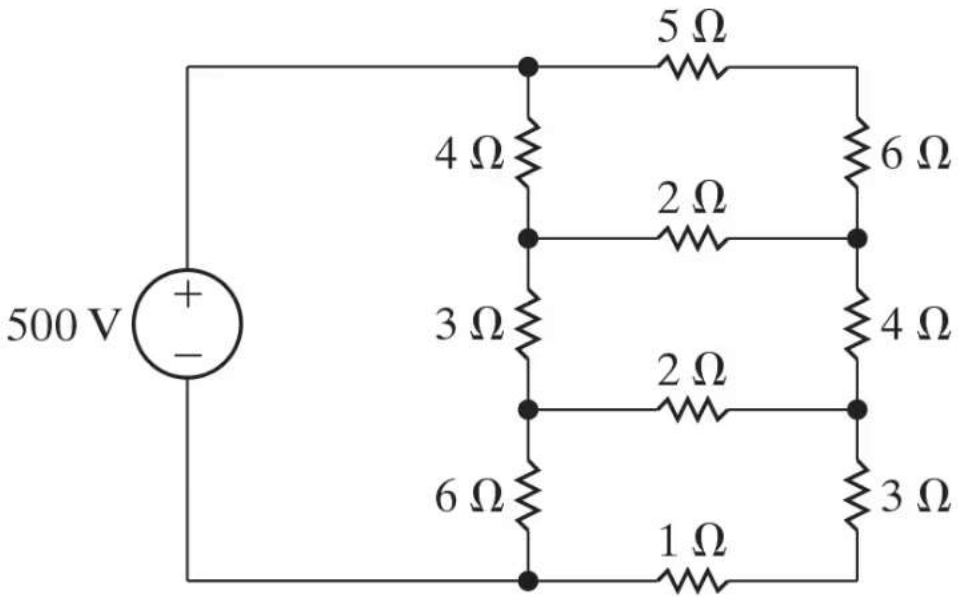
$$v_1\left(\frac{1}{80} + \frac{1}{50} + \frac{1}{40}\right) + v_2\left(-\frac{1}{40}\right) = \frac{50}{80};$$

$$v_1\left(-\frac{1}{40}\right) + v_2\left(\frac{1}{40} + \frac{1}{200} + \frac{1}{800}\right) = 0.75 + \frac{50}{800}.$$

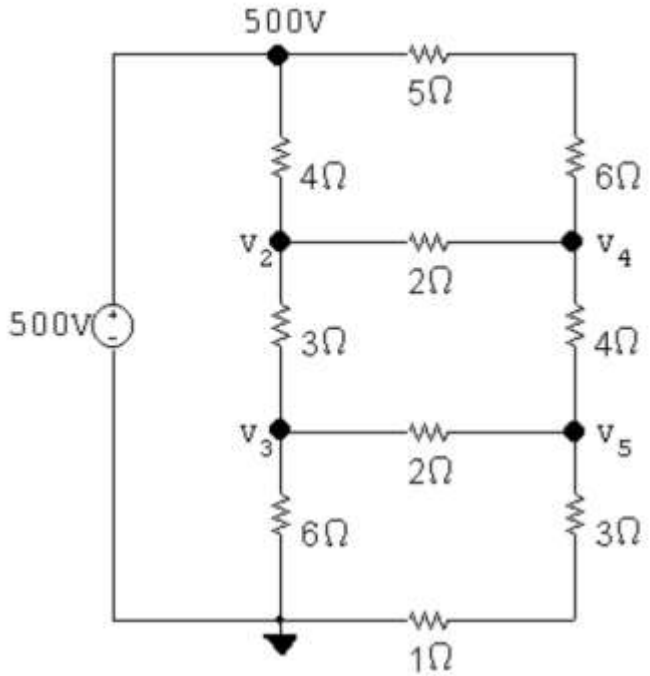
Solving, $v_1 = 34\,\text{V}; \qquad v_2 = 53.2\,\text{V}.$

Thus, $v_o = v_2 - 50 = 53.2 - 50 = 3.2\,\text{V}.$

4.26 **a.** Use the node-voltage method to find the power dissipated in the 5 Ω resistor in the circuit in Fig. P4.26.



4.26 **a.** Use the node-voltage method to find the power dissipated in the 5 Ω resistor in the circuit in Fig. P4.26.



$$\frac{v_2 - 500}{4} + \frac{v_2 - v_4}{2} + \frac{v_2 - v_3}{3} = 0 \quad \text{so} \quad 13v_2 - 4v_3 - 6v_4 + 0v_5 = 1500;$$

$$\frac{v_3 - v_2}{3} + \frac{v_3}{6} + \frac{v_3 - v_5}{2} = 0 \quad \text{so} \quad -2v_2 + 6v_3 + 0v_4 - 3v_5 = 0;$$

$$\frac{v_4 - v_2}{2} + \frac{v_4 - 500}{11} + \frac{v_4 - v_5}{4} = 0 \quad \text{so} \quad -22v_2 + 0v_3 + 37v_4 - 11v_5 = 2000;$$

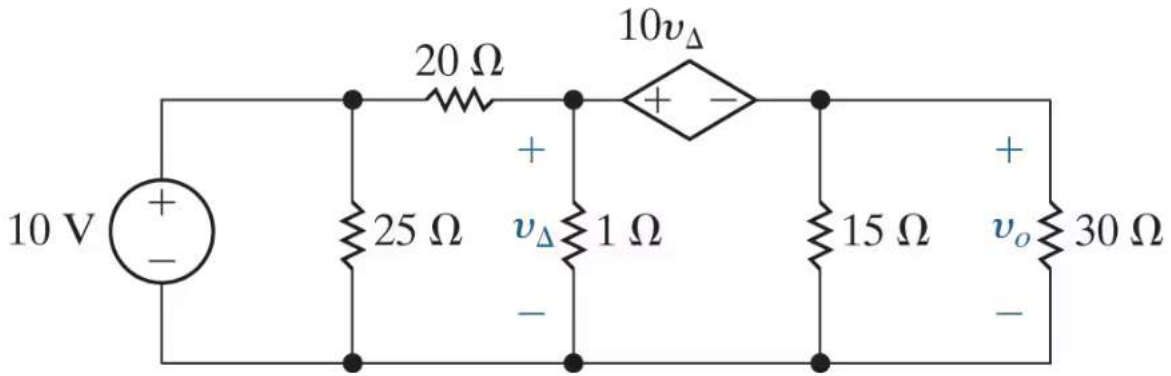
$$\frac{v_5 - v_3}{2} + \frac{v_5}{4} + \frac{v_5 - v_4}{4} = 0 \quad \text{so} \quad 0v_2 - 2v_3 - v_4 + 4v_5 = 0.$$

Solving, $v_2 = 300\,\text{V}$; $v_3 = 180\,\text{V}$; $v_4 = 280\,\text{V}$; $v_5 = 160\,\text{V}$.

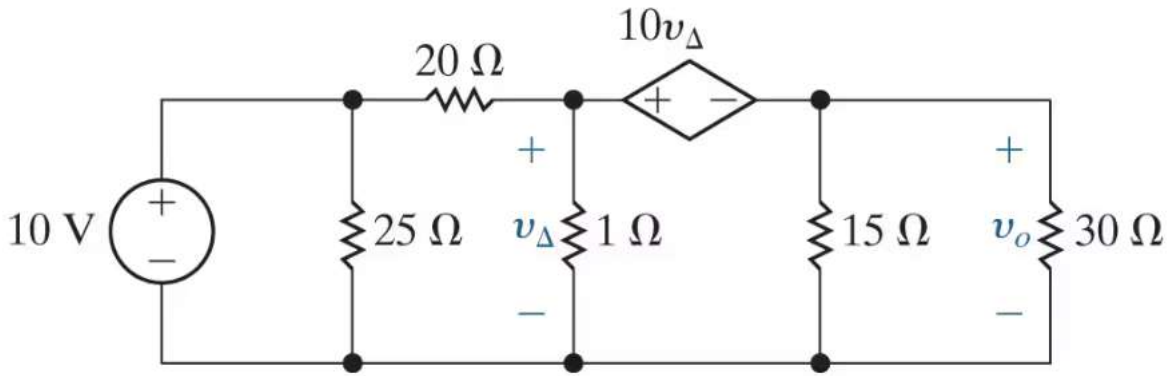
$$i_{5\Omega} = \frac{500 - v_4}{11} = \frac{500 - 280}{11} = 20\,\text{A};$$

$$p_{5\Omega} = (20)^2(5) = 2000\,\text{W}.$$

4.27 Use the node-voltage method to find v_o in the circuit in Fig. P4.27.



4.27 Use the node-voltage method to find v_o in the circuit in Fig. P4.27.



4.27 Placing $10v\Delta$ inside a supernode and using the lower node as a reference, we get

$$\frac{v_\Delta - 10}{20} + \frac{v_\Delta}{1} + \frac{v_\Delta - 10v_\Delta}{15} + \frac{v_\Delta - 10v_\Delta}{30} = 0$$

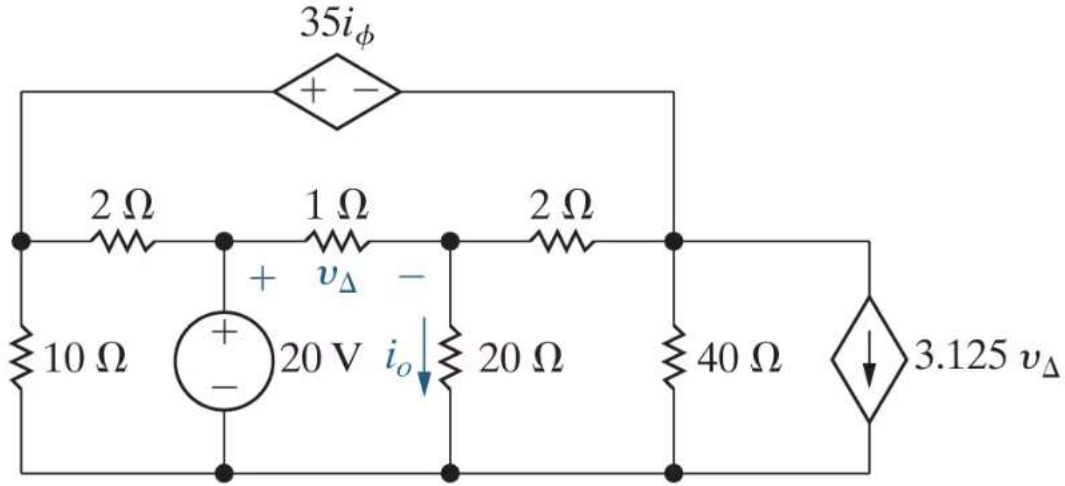
$$\Rightarrow \quad (v_\Delta - 10)3 + 60v_\Delta + (v_\Delta - 10v_\Delta)4 + (v_\Delta - 10v_\Delta)2 = 0$$

$$3v_\Delta - 30 + 60v_\Delta - 36v_\Delta - 18v_\Delta = 0$$

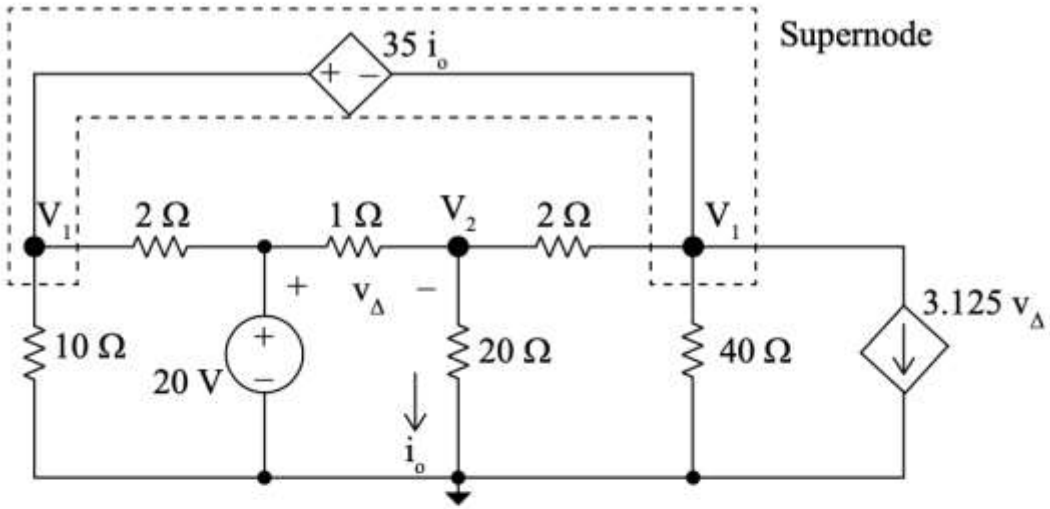
$$9v_\Delta = 30 \Rightarrow v_\Delta = 3.333\,\text{V}$$

$$v_0 = v_\Delta - 10v_\Delta = -9\,(10/3) = -30\,\text{V}$$

4.28 Use the node-voltage method to find the power developed by the 20 V source in the circuit in Fig. P4.28.



4.28 Use the node-voltage method to find the power developed by the 20 V source in the



Node equations:

$$\frac{v_1}{10}+\frac{v_1-20}{2}+\frac{v_3-v_2}{2}+\frac{v_3}{40}+3.125v_\Delta=0;$$

$$\frac{v_2}{20}+\frac{v_2-v_3}{2}+\frac{v_2-20}{1}=0,$$

Constraint equations:

$$v_\Delta=20-v_2;$$

$$v_1-35i_\varphi=v_3;$$

$$i_\varphi=v_2/20.$$

Solving,

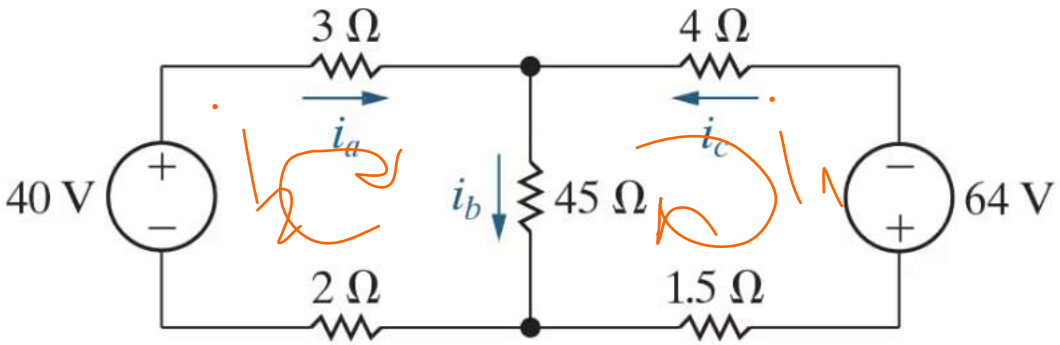
$$v_1=-179.45\text{ V};\quad v_2=-32.88\text{ V};\quad v_3=-121.92\text{ V}.$$

Let i_g be the current delivered by the 20 V source, then

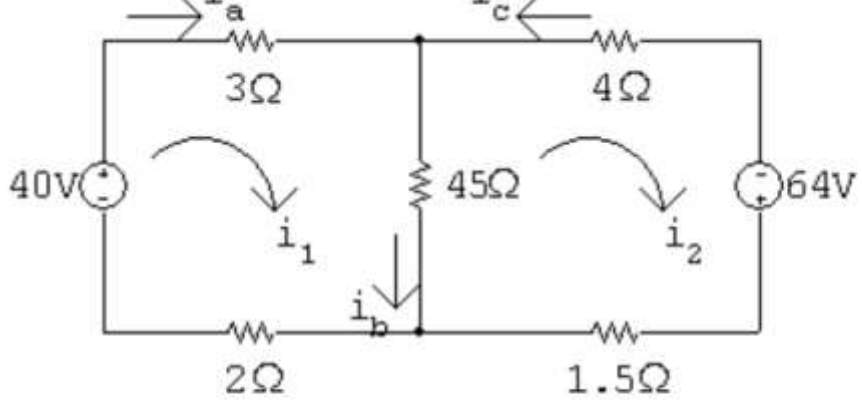
$$i_g=\frac{20-(-179.45)}{2}+\frac{20-(-32.88)}{1}=152.6\text{ A};$$

$$p_g\,(\text{delivered})=20\,(152.6)=3052\text{ W}.$$

4.37 **a.** Use the mesh-current method to find the branch currents i_a, i_b, and i_c in the circuit in Fig. P4.37.



4.37



$$40 = 50i_1 - 45i_2;$$

$$64 = -45i_1 + 50.5i_2.$$

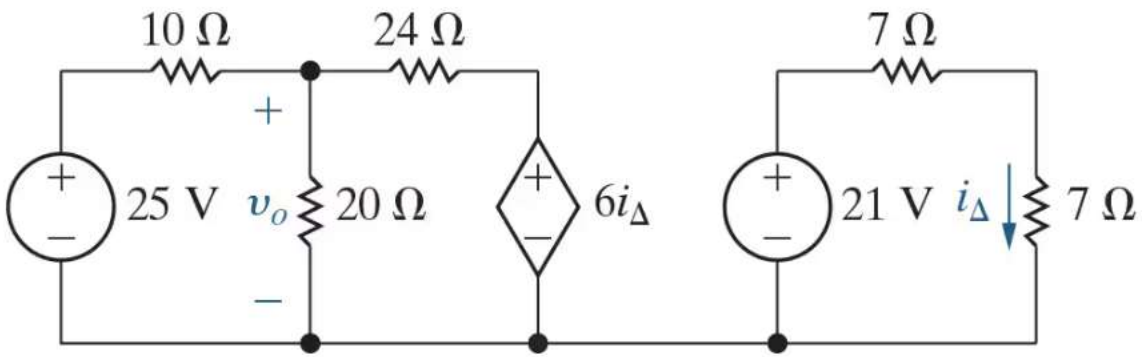
Solving, $i_1 = 9.8\,\text{A}; \quad i_2 = 10\,\text{A}.$

$$i_\text{a} = i_1 = 9.8\,\text{A}; \quad i_\text{b} = i_1 - i_2 = -0.2\,\text{A}; \quad i_\text{c} = -i_2 = -10\,\text{A}.$$

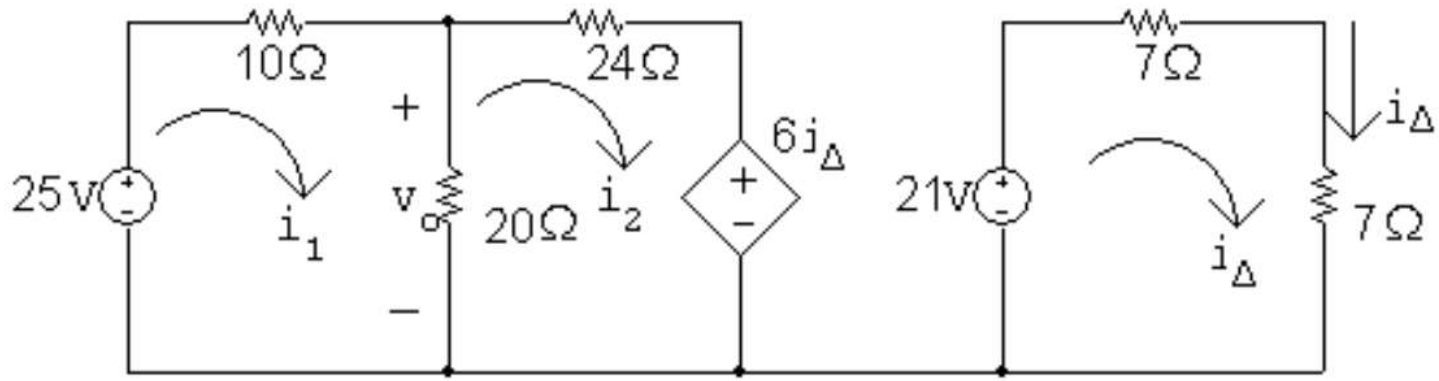
Solving, $i_1 = 9.8\,\text{A}; \quad i_2 = 10\,\text{A}.$

$$i_\text{a} = i_1 = 9.8\,\text{A}; \quad i_\text{b} = i_1 - i_2 = -0.2\,\text{A}; \quad i_\text{c} = -i_2 = -10\,\text{A}.$$

4.39 **a.** Use the mesh-current method to find v_o in the circuit in Fig. P4.39.



4.39 **a.** Use the mesh-current method to find v_o in the circuit in Fig. P4.39.



$$25 = 30i_1 - 20i_2 + 0i_\Delta;$$

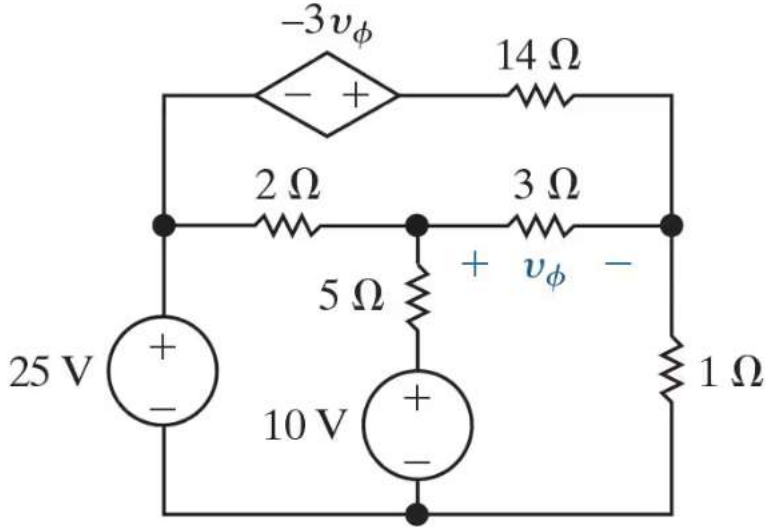
$$0 = -20i_1 + 44i_2 + 6i_\Delta;$$

$$21 = 0i_1 + 0i_2 + 14i_\Delta.$$

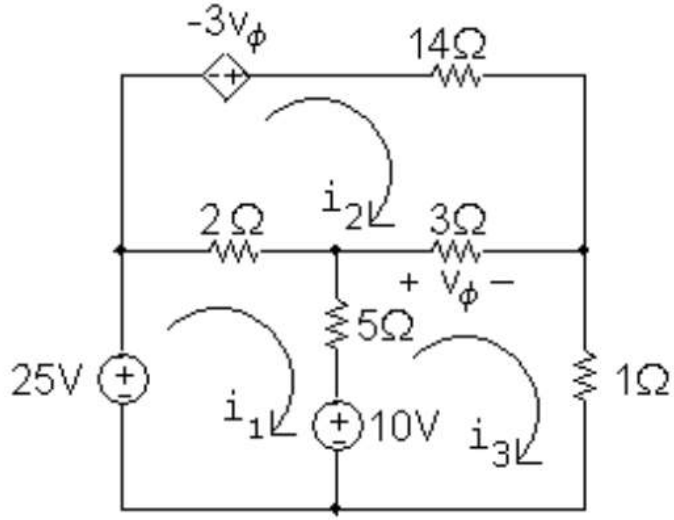
Solving, $i_1 = 1$ A; $i_2 = 0.25$ A; $i_\Delta = 1.5$ A.

$$v_o = 20(i_1 - i_2) = 20(0.75) = 15\,\text{V}.$$

4.41 Use the mesh-current method to find the power developed in the dependent voltage source in the circuit in Fig. P4.41.



Redraw the circuit identifying the three mesh currents:



The three mesh-current equations are

$$\begin{aligned} -25 + 2(i_1 - i_2) + 5(i_1 - i_3) + 10 &= 0; \\ -(-3v_\phi) + 14i_2 + 3(i_2 - i_3) + 2(i_2 - i_1) &= 0; \\ 1i_3 - 10 + 5(i_3 - i_1) + 3(i_3 - i_2) &= 0. \end{aligned}$$

The dependent source constraint equation is

$$v_\phi = 3(i_3 - i_2).$$

Place these four equations in standard form:

$$\begin{aligned} 7i_1 - 2i_2 - 5i_3 + 0v_\phi &= 15; \\ -2i_1 + 19i_2 - 3i_3 + 3v_\phi &= 0; \\ -5i_1 - 3i_2 + 9i_3 + 0v_\phi &= 10; \\ 0i_1 + 3i_2 - 3i_3 + 1v_\phi &= 0. \end{aligned}$$

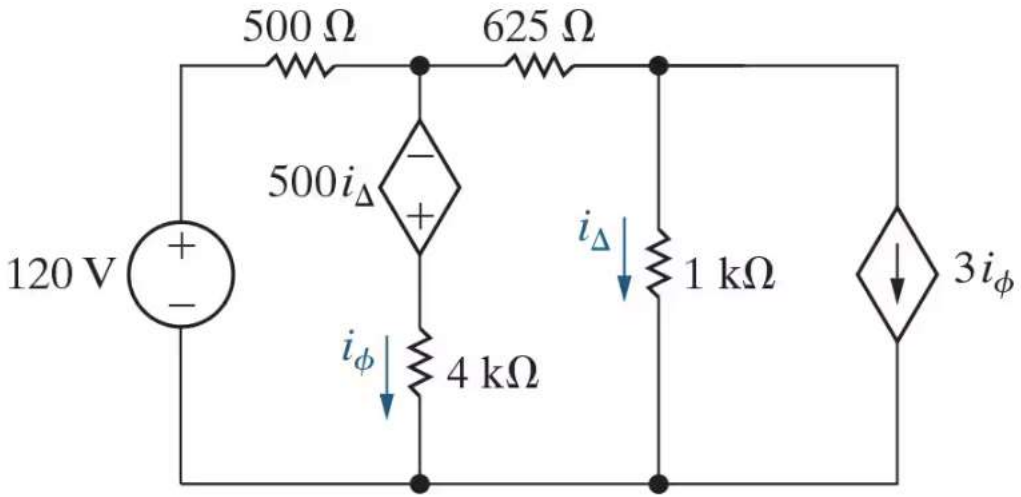
Solving,

$$i_1 = 4\,\text{A}; \qquad i_2 = -1\,\text{A}; \qquad i_3 = 3\,\text{A}; \qquad v_\phi = 12\,\text{V}.$$

$$p_{\text{ds}} = -(-3v_\phi)i_2 = 3(12)(-1) = -36\,\text{W}.$$

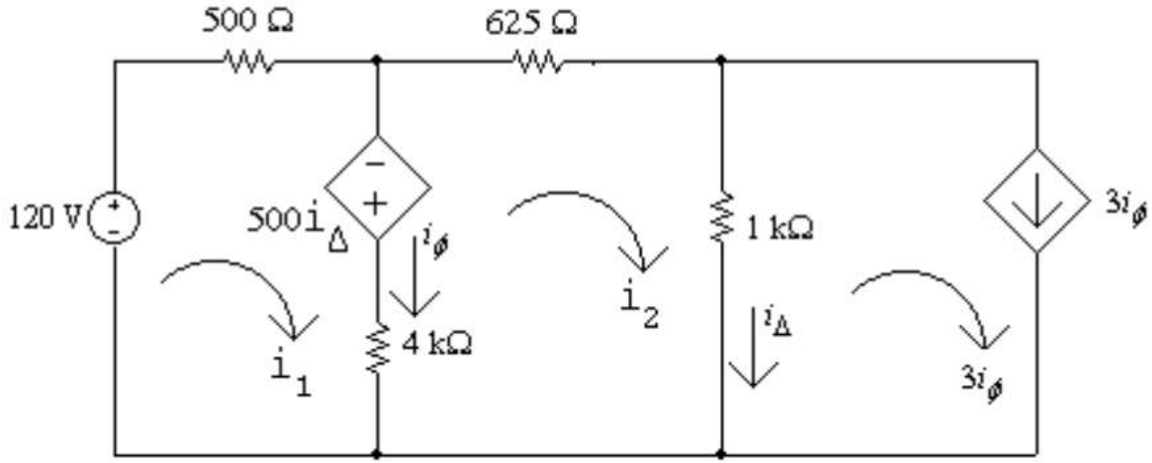
Thus, the dependent source is delivering 36 W.

4.48 **a.** Use the mesh-current method to determine which sources in the circuit in Fig. P4.48 are generating power.



4.48 **a.** Use the mesh-current method to determine which sources in the circuit in Fig. P4.48 are generating power.

P 4.48



Mesh current equations:

$$-120 + 500i_1 - 500i_\Delta + 4000(i_1 - i_2) = 0;$$

$$625i_2 + 1000(i_2 - 3i_\phi) + 4000(i_2 - i_1) + 500i_\Delta = 0.$$

Dependent source constraint equations:

$$i_\Delta = i_2 - 3i_\phi;$$

$$i_\phi = i_1 - i_2.$$

In standard form:

$$4500i_1 - 4000i_2 - 500i_\Delta + 0i_\phi = 120;$$

$$-4000i_1 + 5625i_2 + 500i_\Delta - 3000i_\phi = 0;$$

$$0i_1 - 1i_2 + 1i_\Delta + 3i_\phi = 0;$$

$$-1i_1 + 1i_2 + 0i_\Delta + 1i_\phi = 0.$$

4.48 **a.** Use the mesh-current method to determine which sources in the circuit in Fig. P4.48 are generating power.

Solving,

$$i_1 = 100\,\text{mA}; \qquad i_2 = 80\,\text{mA}; \qquad i_\Delta = 20\,\text{mA}; \qquad i_\phi = 20\,\text{mA}.$$

[a] Source power:

$$p_{120\text{V}} = -120 i_1 = -120(0.1) = -12\,\text{W};$$

$$p_{\text{CCVS}} = -500 i_\Delta i_\phi = -500(0.02)(0.02) = -0.2\,\text{W};$$

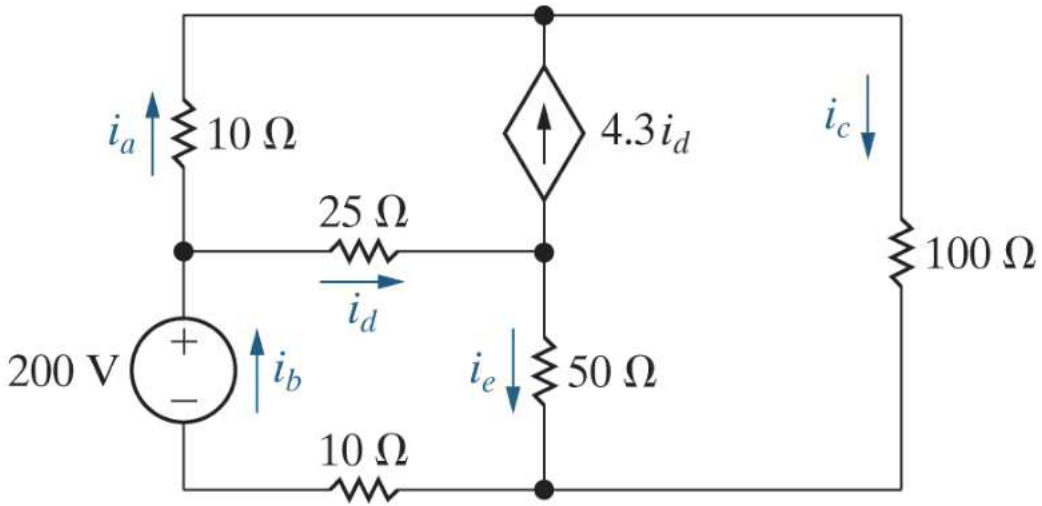
$$v_{\text{CCCS}} = 1000 i_\Delta = 1000(0.02) = 20\,\text{V};$$

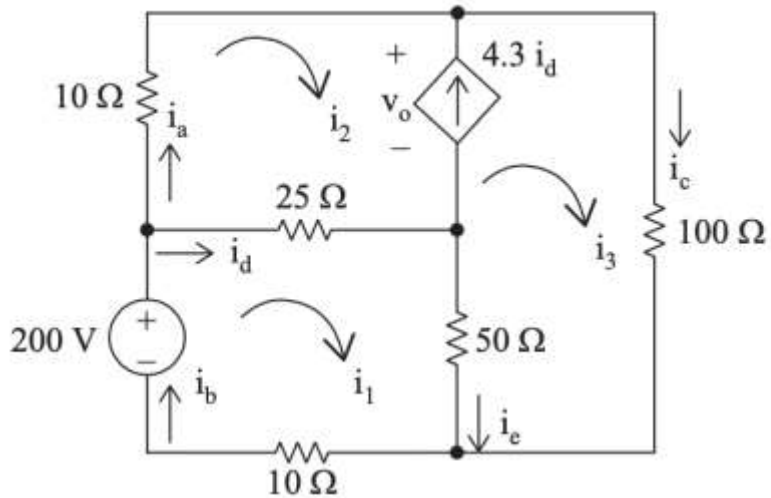
$$p_{\text{CCCS}} = 3 i_\phi v_{\text{CCCS}} = 3(0.02)(20) = 1.2\,\text{W}.$$

Thus the two voltage sources are generating power and

$$p_{\text{generated}} = 12 + .2 = 12.2\,\text{W}.$$

4.50 **a.** Use the mesh-current method to find all the branch currents in $i_a - i_e$ in the circuit in Fig. P4.50.





$$200 = 85i_1 - 25i_2 - 50i_3$$

$$0 = -75i_1 + 35i_2 + 150i_3 \qquad \text{(supermesh)}$$

$$i_3 - i_2 = 4.3(i_1 - i_2)$$

Solving, $i_1 = 4.6$ A; $\quad i_2 = 5.7$ A; $\quad i_3 = 0.97$ A

$$i_\mathrm{a} = i_2 = 5.7 \text{ A}; \qquad i_\mathrm{b} = i_1 = 4.6 \text{ A}$$

$$i_\mathrm{c} = i_3 = 0.97 \text{ A}; \qquad i_\mathrm{d} = i_1 - i_2 = -1.1 \text{ A}$$

$$i_\mathrm{e} = i_1 - i_3 = 3.63 \text{ A}$$